On the discussion questions about Tarski's 'The Semantic Conception of Truth' (November 19, 2054)

Here are a couple of thoughts and observations on the discussion questions for your perusal. Please do not hesitate to comment, question, object, expand, or otherwise react to them.

All groups:
"Suppose we are convinced that truth for natural languages cannot be defined in the same way as is done for formal languages. What would be the consequence for how we view the relationship between logic and philosophy?"

Tarski's paper in fact contains two reasons for doubting that truth can be defined for a natural language. The first reason is that natural languages are semantically closed, i.e., they contain their own truth predicate and allow for sentences that are self-referential. But note that actually this is not so much an objection against the possibility of defining truth fur such languages, but a consideration about the usefulness of doing so. Truth *can* be defined, also for semantically closed languages, but it leads to semantic paradoxes. And the latter consequence is what, in the eyes of Tarski, makes the entire enterprise of defining truth for semantically closed languages pointless More on that below, in the considerations regarding question 1.

The second reason that Tarski gives is that a definition of truth for a language *L* requires that *L* 'have a specified structure', i.e., that it allows a completely explicit definition of its well-formed expressions, including of its sentences. Natural languages, Tarski claims, do not have such a specified structure. He does not give much argument for this claim, but in making it he echoes what was the *communis opinio* in much of philosophy at the time. As we have also seen in the texts of Frege and Russell that we read at the beginning of the course, the conviction that natural language is unsystematic, vague, ambiguous and lacks precision lies at the root of the view that we need formal tools, such as logic and mathematics, to conduct proper investigations, not just in science, but also in philosophy.[1] This view has informed logical positivism and the views of philosophers such as Quine.

It was only with the rise of generative grammar in the late 1950s and early 1960s that this, slowly, started to change. Noam Chomsky started to apply what were basically concepts from mathematical logic in the study of the syntax of natural languages, and then expanded and modified the resulting frameworks over the years. What it showed to philosophers was that natural languages did have a 'specified structure' after all, and therefore that at least *that* objection to applying something akin to Tarski's definition of truth to them actually was not really valid. That started the development of formal semantics, where concepts and techniques from logic, mathematics and computer science are applied in the study of natural language meaning. One of the pioneers of formal semantics, Richard Montague, who was a student of Tarski, formulated this fundamental change in perspective as follows in his 1970 paper 'Universal Grammar':

> There is in my opinion no important theoretical difference between natural languages and the artificial languages of logicians; indeed, I consider it possible to comprehend the syntax and semantics of both kinds of languages within a single natural and mathematically precise theory. On this point I differ from a number of philosophers, but agree, I believe, with Chomsky and his associates.

[1] As a matter of fact, the idea that natural language may lead us astray is a view that we find not just in (early) analytic philosophy, but through the ages and in many different traditions.

But if the second reason to think that truth cannot be defined for natural languages is out of way, what about the first one? One could say: yes, we *can* do it, but should we, given the fact that we will have to face paradoxes? The answer to this question is not that often discussed in the formal semantics literature. Donald Davidson, one of the other pioneers, is an exception. In his 1967 paper 'Truth and Meaning' he states:

> The semantic paradoxes arise when the range of the quantifiers in the object language is too generous in certain ways. But it is not really clear how unfair to Urdu or to Wendish it would be to view the range of their quantifiers as insufficient to yield an explicit definition of 'true-in-Urdu' or 'true-in-Wendish'. Or, to put the matter in another, if not more serious, way, there may in the nature of the case always be something we grasp in understanding the language of another (the concept of truth) that we cannot communicate to him. In any case, most of the problems of general philosophical interest arise within a fragment of the relevant natural language that may be conceived as containing very little set theory. Of course, these comments do not meet the claim that natural languages are universal. But it seems to me that this claim, now that we know such universality leads to paradox, is suspect.

Clearly, Davidson chooses for the option of restricting the segment of the language that we define truth for, assuming that what we need in terms of natural language resources to do philosophy is not hampered by that restriction. Whether that actually works is a complicated question, that we cannot go into here further.[2]

But one might also choose the other horn of the dilemma and accept that natural language by their very nature allow for the formulation of semantic paradoxes and that such is a fact that any comprehensive semantic theory of a natural language needs to account for. Then we acknowledge that natural languages and the formal languages of logic and mathematics differ in an essential way. Where consistency is a key requirement of the latter,[3] it is one that the former simply do not meet.

Now can one do philosophy using natural language, given that the latter is inconsistent? One can try to answer this question in a principled manner, in the way that Frege and Russell and Tarski and many others have done. Or one could look at the matter from a more practical point of view: How do we do philosophy? What ways of doing philosophy are useful? It is not entirely clear that this way of approaching the questions delivers an uncontested answer, but what is clear is that most philosophical work is done in natural language and that the alleged inconsistency of the latter seems to play no real role in the choice of philosophers to do so.

Group 1:
"Natural languages appear to be semantically closed. Does that mean that it is impossible to formulate a truth definition for them? Is the fact that we can formulate paradoxes in natural language an obstacle?"

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[2] For some in-depth discussion, see Ernie Lepore & Kirk Ludwig's 'The problem of semantic defects in natural languages', which is chapter 10 in their book *Donald Davidson. Meaning, Truth, Language, and Reality*, OUP, 2005.

[3] There is a small number of logicians that reject this and that study so-called 'paraconsistent' logics, in which the occurrence of contradictions does not have the disastrous effects it has in classical logic due to the EFSQ-rule that the latter embrace. Paraconsistent logic is often associated with so-called 'dialetheism', the view that there are true contradictions (such as Russell's paradox or the Liar-paradox); its inspiration comes also from studying system of thought such as certain forms of Buddhist logic, such as the *catuskoti* that figures prominently in Nagarjuna's writings (see, e.g., Graham Priest, 'The logic of the *catuskoti*', in *Comparative Philosophy*, 2010, vol. 1(2), pp. 24-54. More generally, paraconsistent logics are motivated by the fact that in real life people often must reason with inconsistent information. A logic, such as classical logic or intuitionistic logic, that does not allow for inconsistencies is then of no use.

That natural languages are semantically closed in the sense in which Tarski defines that concept (viz., that they allow for self-reference and contain their own truth-predicate) seems not something we could question. And that indeed means that semantic paradoxes, such as the Liar-paradox, can be formulated. The question then is what consequences this has.

One view is that the problems with semantically closed languages can be worked around, provided we are willing to accept that we cannot define truth for the entire language. That is, we restrict the definition of truth to a proper subpart of the language, i.e., the subset of all of its well-formed sentences that do not contain the predicate "is true" and/or that do not refer to themselves. The latter is more easily said than done, as self-reference can sneak in indirectly:

> *A*: What *B* will say is not true
> B*:* What *A* just said is true

Preventing such indirect cases of self-reference is tricky. Obviously, we cannot rule such sentences out since there are perfectly innocent occurrences as well:

> *A*: What *B* will say is not true
> B*:* Rob Jetten will be the next prime minister of the Netherlands
>
> Or:
>
> *A*: Beijing has more than 15 million inhabitants
> B*:* What *A* just said is true

So, individually these sentences are unproblematic, it is only in particular combinations that they lead to a paradox. In view of that the best strategy seems to be to simply take into consideration only that subset of sentences of a given natural language that do not contain the truth predicate of that language. (There is no problem with sentences of NL_1 that contain the truth predicate for another language NL_2 and that talk about the sentences of NL_2, as, e.g., in: 'The sentence "Macron est le président de la France" is a sentence that is $\text{true}_{\text{in French}}$.')

Another view is to accept paradoxes that result from the language being semantically closed. Here the key thing to note is that defining truth is one thing but a truth definition serving as the foundation for a logic is another. In the case of semantically closed languages, we can define truth in the same way as for languages that are not semantically closed. And that definition can be (indeed, should be) materially and formally adequate. But it will lead to inconsistencies and that is a problem, as long as we require a logical system to be consistent.[4] For more on that, cf., above, in connection with the "all groups" question.

Group 2:
"The redundancy view claims that 'It is true that *p*' and '*p'* says the same thing, and that therefore the 'It is true that'-part can be eliminated. Tarski argues against that by noting that one cannot eliminate 'true' from, e.g., 'The first sentence written by Plato is true'. Discuss. Are both sides talking about the same kind of use of 'true'?"

[4] Cf, footnote 3 above about possible alternatives.

It appears that when we say something like 'What she said is true' what we say is about what she said. *We* say this in order to substantiate what *she* said, so we say it to confirm, to support, to take sides. None of that is what we do, or can do, when it comes to what we say ourselves. The only function of *us* saying 'It is true that *p*' instead of just saying '*p*' seems to be to emphasise, or to allay doubts, or what not. We say '*p*', someone else questions that ('Really? *p?*'), or even claims the opposite ('No, not-*p*!'). Then we retort: 'It is true that *p*'. Or, different case, we use it to formulate a concession, only to come back with another claim: 'Yes, it is true that *p*, but don't forget that *q*, and that's much more important!'

So, there are a number of different uses for the phrase 'It is true that *p*', and they are all different from what we use '*p*' for. However, in all these cases of our saying 'It is true that *p*' what we *claim* is the same as what we claim when we just say '*p*'. We commit ourselves to defending that *p* is the case.[5] That seems to support the redundancy view: in terms of the claim that we make 'It is true that *p*' does not differ from '*p*', it just signals something about the context in which we state that claim.

But what about Tarski's examples? As was noticed above, it seems that the use of 'is true' in an example such as 'What she said is true' is different from the use of 'It is true that' in the examples adduced by proponents of the redundancy view. In such examples we refer to what was said by someone else, using a description, viz., 'What she said', or 'The first sentence written by Plato'. Now the crucial difference between something like ' "Snow is white" is true' and 'What she said is true' is that the first uses a particular kind of name and the second a description. The name of a sentence that is formed by putting the sentence between quotation marks is directly connected to that sentence: we can *see* what sentence is named, there can be no doubt about what the name refers to.[6] With the description 'What she said' this is different: by itself it does not identify a particular sentence, we need additional information to determine its reference. In a case like this that depends heavily on the context: she probably said more than one thing, so do you mean what she said just now? Or last night, when the two of you were talking on the phone? But even when context does not play a role, as in Tarski's example 'The first sentence written by Plato', we need additional information to determine what the description refers to. But suppose we would know what that sentence is, say it is 'Athens is a great city'. Can we then replace 'The first sentence written by Plato' by ' "Athens is a great city" '? I.e., are 'The first sentence written by Plato is true' and ' "Athens is a great city" is true' equivalent? The answer seems to be that they are. But they do not have the same meaning, for what sentence was the first one that Plato wrote surely is a contingent matter.

Group 3:
"The pragmatist view of truth is that 'True is what works'. Is that necessarily at odds with Tarski's semantic conception? If so, where do the two clash? If not, how can you combine them?"

It is important to bear in mind that Tarski explicitly claim that his aim is *not* to propose an epistemological theory. His semantic theory of truth does not make any claims about when we are warranted to maintain that something is true, or when we can take it to be decisively refuted, nor does it tell us what ways there are for finding out whether something is true or not, let alone what is the best way to go about doing that. It is called a 'semantic theory of truth', because that

[5] That meaning can be explained in terms of commitments is the fundamental idea that underlies Robert Brandom's approach, as first worked out in his *Making It Explicit* (1994).
[6] I.e., quotation forms names that are so-called 'rigid designators'. We will encounter that concept when we read Kripke's paper 'Identity and Necessity' later in the course.

is what it is: it tells us what 'is true-in-*L*' means. And it is an extensional theory: it specifies the meaning of 'is true-in-*L*' by laying out the extension of this predicate. In doing that, it does not tell us which sentences of *L* actually are true or not, it does not specify truth *values*, it states truth *conditions*.

The pragmatist conception of truth is rather on the opposite of this spectrum: it aims to give a criterion for when to accept something as true. Certain correspondence theories also do that, but in a different way. Thus, a theory that states that a sentence is true if and only if it describes an existing state of affairs, also gives us a criterion for accepting a sentence: check what state of affairs it describes and then check whether that state of affairs does indeed exist. In such a case the criterion can be said to be 'ontological': truth is directly related to what the world is like. The pragmatist conception also gives us a criterion, but one that is 'practical', rather than ontological. It refrains from referring to what the world is like and does so because it maintains that whether there is such a thing as 'what the world is like' is beyond our grasp. We can devise theories and test them, and when they work, we accept them as true. That acceptance is always only temporary: with new investigations, new methods, new tools, we may find out that another theory works better and then the rational thing to do is to accept that one as 'true'.

Viewed in this way, there seems to be no *principled* reason why Tarski semantic conception of truth could not be combined with a pragmatist conception. The first describes what 'is true' means, what the logic of the concept of truth is. The second deals with an entirely different matter: when is it rational for us to accept a sentence, or a theory, as true. The first does pose some constraint on the second: the semantic conception of truth forms the basis of a conception of logical consequence. So, if we would want to combine the semantic conception with a pragmatist one, we need to assume that the latter's conception of rationality follows the rules laid out by the former. There may be reasons not to do that, and in that case the combination would not be possible. But those reasons will be particular, not principled ones.

Group 4:
"On page 360 Tarski observes that when asked the majority of people agree with a formulation that corresponds to the semantic conception of truth, while only a minority do so with a correspondence formulation. Do you agree that this type of empirical data is relevant or philosophical questions? If so, why? If not, why not?"

Historical note: the work that Tarski refers to was done by the Norwegian philosopher and logician (and later environmental activist) Arne Naess, who reported the results in his 1938 paper ' "Truth" as Conceived by Those Who Are Not Professional Philosophers'. Naess, who called his approach 'experimental philosophy', did not create a large following and his work was largely forgotten in the second half of the 20th century. It was only in the early 2000s that the idea of doing philosophy by making use of empirical data has been taken up again.

Traditionally, much philosophy, certainly in the analytic tradition, depends on intuitions. Philosophers contemplate a certain problem, think up specific, concrete cases that illustrate various aspects of the problem, develop intuitions about what those cases imply, and then formulate an answer to the initial problem. That methodology is used widely, from moral philosophy ('Is being blameworthy a good criterion for attributing moral responsibility?') to philosophy of mind ('Is it possible for a human to experience the world the way in which a bat does?') to philosophy of language ('Do definite descriptions determine reference in the same way as proper names?'). Of course, it's not just intuitions that make up a (good) philosophical analysis: intuitions need to be justified and that means that it needs to be argued that they lead to

consistent and insightful analyses. That involves coming up with logically sound and convincing arguments, which is not a matter of intuition. But intuitions are usually what such analyses start from and what they are tested with.

The reliance on intuitions has been questioned every now and then[7], but the development of an alternative method came about only later, with the work of Knobe, Prinz, Stich, Machery, and others.[8] They started to collect empirical data, by means of questionnaires and by doing controlled experiments, consulting 'ordinary' people (i.e., people who are not professional philosophers) on their intuitions about philosophical issues such as whether there is free will, what is the nature of the mind/consciousness, or the nature of knowledge (including the so-called 'Gettier-cases'), and so on. Socio-cultural variation on these issues was also studied extensively.

To what extent and under which conditions is this type of empirical data is relevant for more traditional philosophical analysis? Note that the topics themselves are philosophical, in the usual way. It is the data that are different. Two questions (minimally) arise here. First, is there any reason to assign the same weight to the intuitions of non-professionals and professionals? And second, what are we to make of the differences between people's intuitions?

The first question really turns on the specialized expertise of professional philosophers. It seems reasonable (though certainly not always unquestionable) that professional philosophers are more skilled in analysing complex problems, tracing the consequences of certain views, spotting inconsistencies, and noticing and exploring connections with other problems and views. But that is about what comes *after* the basic intuitions. It would seem odd to say that philosophers have better intuitions just because they happen to be philosophers. Unless, of course, we concede that a philosopher's intuitions are nurtured by their philosophical knowledge and analytic skills. But then it is not at all straightforward how we are to compare the 'raw' intuitions of non-philosophers with the theoretically informed intuitions of philosophers. In fact, it seems questionable whether the two are intuitions in the same sense, despite both being called intuitions.

The second question concerns variation in the data. Where there is great consensus, it seems that variation may be safely ignored. If 96% of the people agree that moral responsibility induces blameworthiness, it seems reasonable to leave the disagreeing 4% out of consideration. But what if we have a philosophical issue that gives rise to a more substantial split, say 65% versus 35%? As an empirical data point that is just what it is: 65% go with view *A*, and 35% go with view *B*. If we are conducting an empirical investigation, and are interested in say, correlations between holding view *A* or view *B*, and political affiliation, or socio-economic status, this is just data. But if we want to use such data in a philosophical analysis of problem for which these two views represent competing solutions, we are in trouble. It seems we cannot simply go with the majority view. Rather, it seems we must accept that what we thought was one problem, with competing solutions, isn't that. But that means that the very conception of the problem is no longer a philosophical matter.

[7] See for example, Jaakko Hintikka's 1999 paper 'The Emperor's New Intuitions'.
[8] For an overview, see the SEP entry on experimental philosophy by Joshua Knobe and Shaun Nichols (https://seop.illc.uva.nl/entries/experimental-philosophy/).